

Agent-OM: Leveraging LLM Agents for Ontology Matching

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Introduction

Leveraging LLMs for downstream tasks has several challenges:

- Do not capture late-breaking information
- LLM hallucinations hamper reliability
- Limited capabilities in non-linguistic tasks

Why do we need LLM agents for OM?

- Planning $\Rightarrow$ Chain of Thought (CoT)
- Tool Use $\Rightarrow$ Function Calling (FC)
- Short-term Memory $\Rightarrow$ In-context Learning (ICL)
- Long-term Memory $\Rightarrow$ Retrieval-Augmented Generation (RAG)

Key Features:

- A LLM acts as a central “brain” to link different modules and instruct their functions via prompt engineering
- A pair of planning modules use CoT for OM decomposition
- A set of OM tools use ICL/RAG to mitigate LLM hallucinations
- A shared memory module uses dialogue and hybrid data storage to support the search and retrieval of entity mappings

Advances:

- Cost-effective entity information retrieval
- Matching candidate selection
- Search-based matching functions

Performance:

- Very close to the best long-standing on simple OM tasks.
- Significant improvements on complex and few-shot OM tasks.

Related Work

- Knowledge-based OM systems [10, 11, 5, 4, 19, 12]
- Machine learning (ML)-based OM systems [7, 3]
- Early studies using LLMs for OM [8, 14, 9, 6, 1]
- Using agents for ontology-related tasks [17, 2, 15, 18]

Matching with LLM Agents

Matching with LLM Agents

A classical matching process:

$$R_{int} \Rightarrow R_{ext} \Rightarrow M_{sel} \Rightarrow M_{alg} \Rightarrow M_{ref} \quad (1)$$

An agent-based matching process:

$$Agent_R(R_{int} \Rightarrow R_{ext} \Rightarrow R_{sto}) \Rightarrow Agent_M(M_{sea} \Rightarrow M_{sel} \Rightarrow M_{alg} \Rightarrow M_{ref}) \quad (2)$$

Abbreviations:

- R_{int} = Internal retriever
- R_{ext} = External retriever
- R_{sto} = Store information
- M_{sea} = Search information
- M_{sel} = Candidate selection
- M_{alg} = Matching algorithm
- M_{ref} = Matching refinement

Matching with LLM Agents - Continued

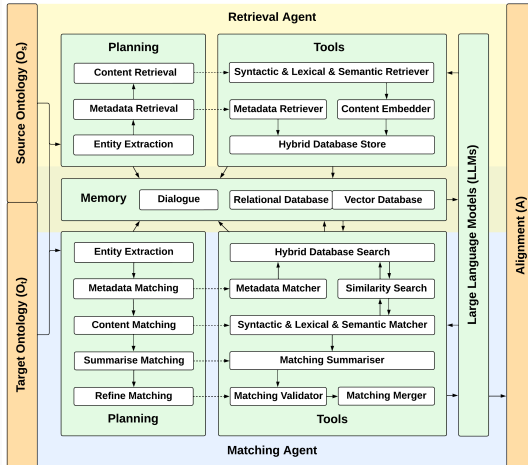


Figure 1: Architecture of Agent-OM. All components are executed twice, once for each of the source and target ontologies, apart from the matching merger tool that combines the results from each pass.

Evaluation

Evaluation - OAEI Conference Track

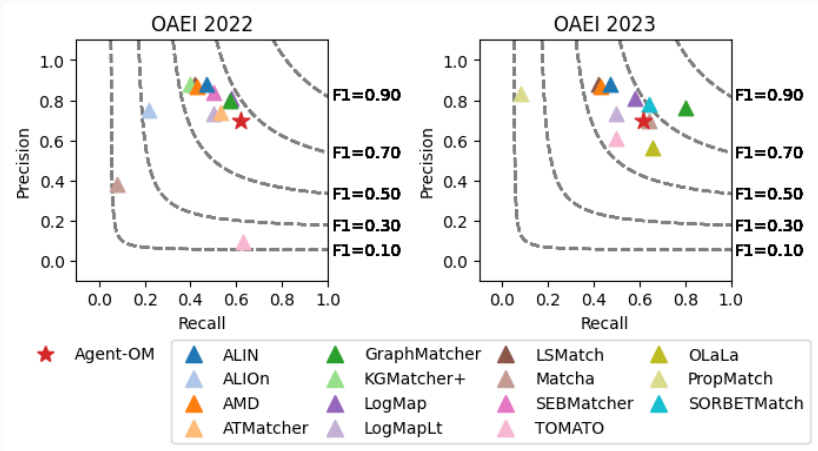


Figure 2: OAEI Conference Track Test Case.

Evaluation - OAEI Anatomy Track - Test Case 1

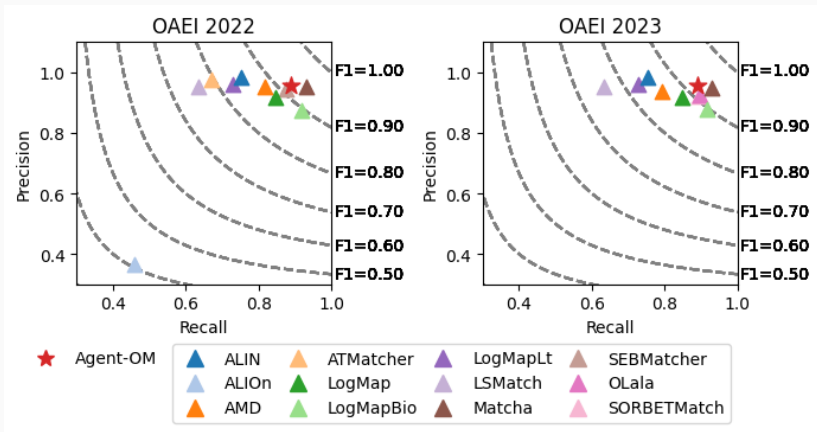


Figure 3: OAEI Anatomy Track Test Case 1.

Evaluation - OAEI Anatomy Track - Test Case 2

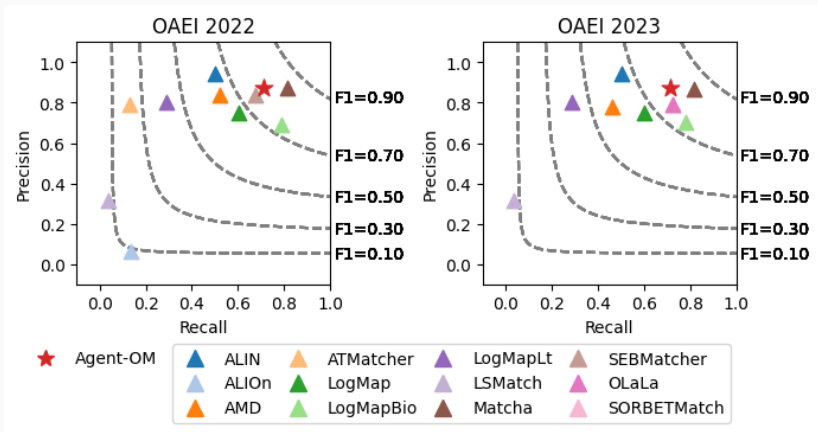


Figure 4: OAEI Anatomy Track Test Case 2.

Evaluation - OAEI MSE Track - Test Case 1

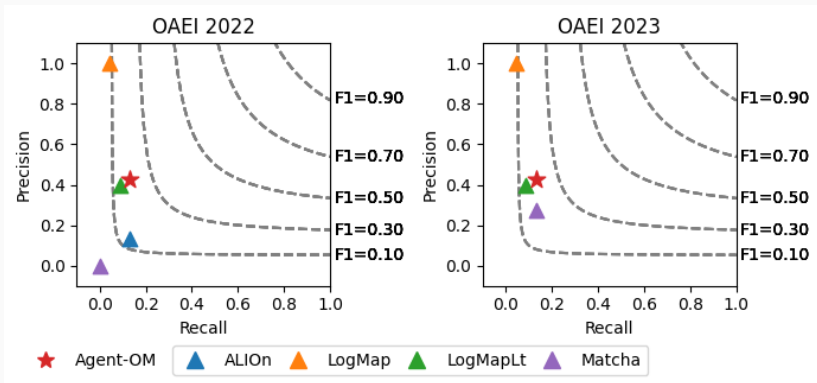


Figure 5: OAEI MSE Track Test Case 1.

Evaluation - OAEI MSE Track - Test Case 2

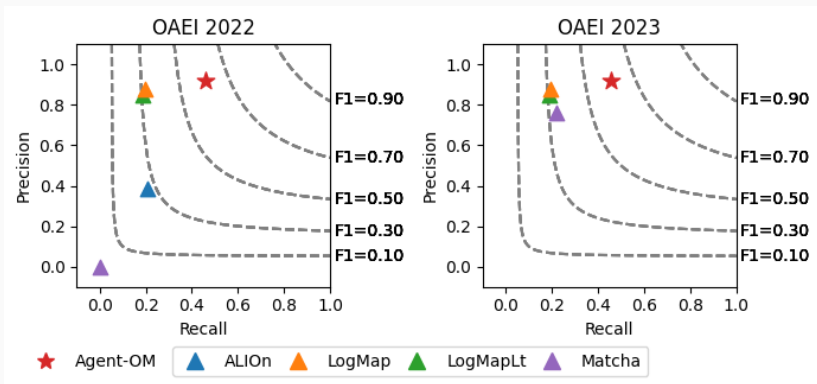


Figure 6: OAEI MSE Track Test Case 2.

Evaluation - OAEI MSE Track - Test Case 3

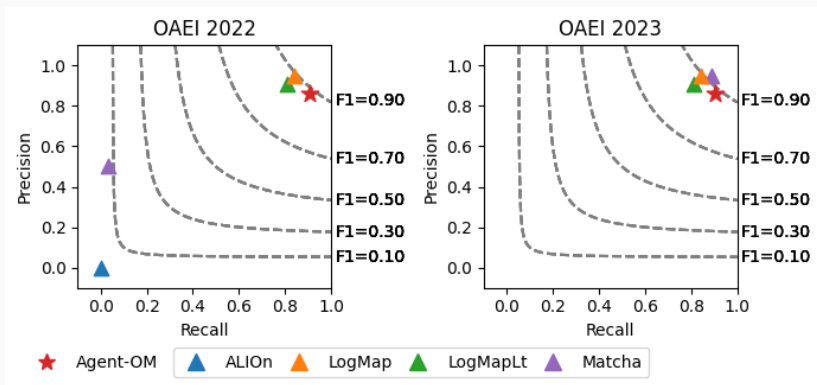


Figure 7: OAEI MSE Track Test Case 3.

Ablation Study

Ablation Study - Comparison of LLM-based architectures

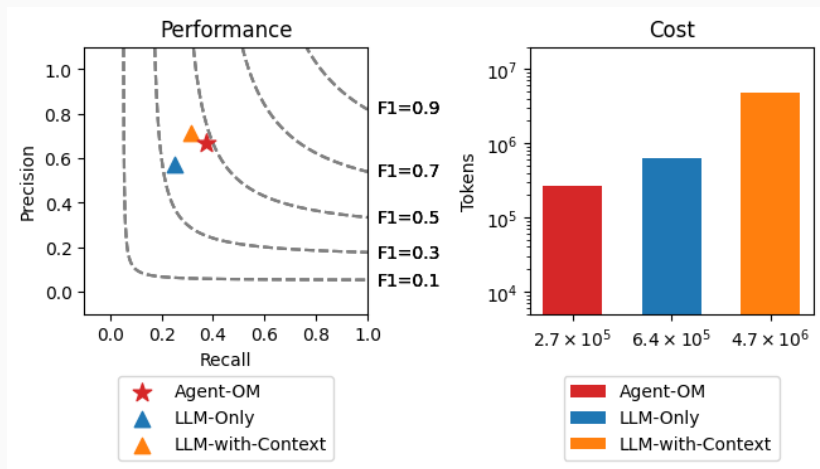


Figure 8: Comparison with LLM-based architectures.

Ablation Study - Comparison of different LLMs

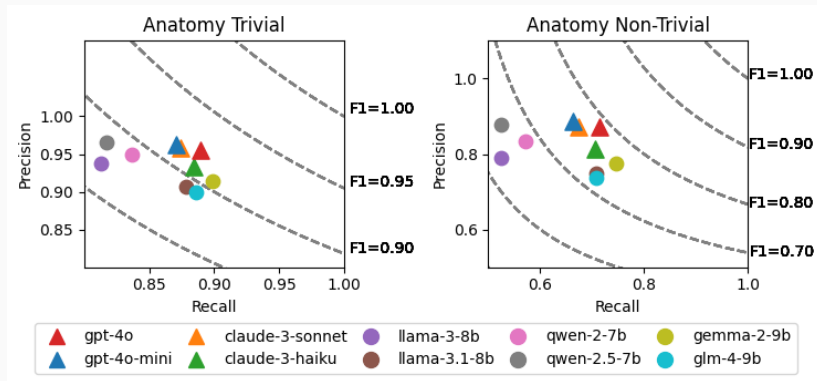


Figure 9: Comparison of different LLMs. API-accessed models are shown as triangles and open-source models as circles.

Ablation Study - Matching Validator

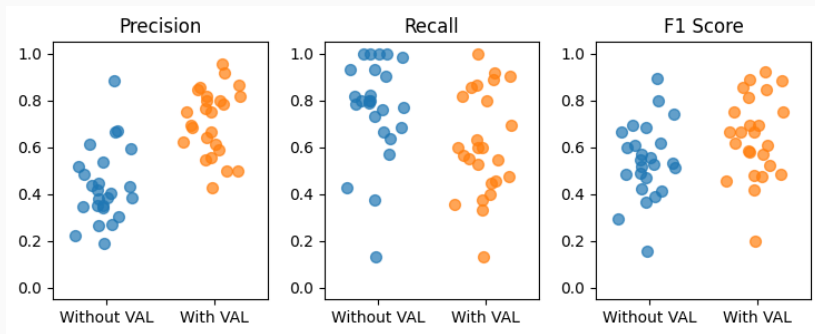


Figure 10: Comparison of without/with matching validator.

Ablation Study - Matching Merger

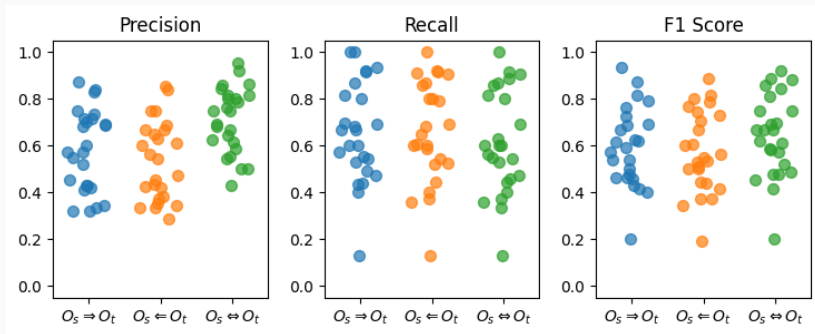


Figure 11: Comparison of $O_s \Rightarrow O_t$, $O_s \Leftarrow O_t$, and $O_s \Leftrightarrow O_t$

Discussion

Findings:

- More efficient and effective than LLM-based OM.
- Context learning, transitive reasoning, and self correction.

Implications:

- A matching process could be more complex.
- Prompt engineering is the key to instructing efficient LLM agents.
- LLM hallucinations can be mitigated, but cannot be eliminated.
- There is a trade-off between precision and recall.
- For LLMs used for OM, we find Moravec's paradox [13]: “the hard problems are easy and the easy problems are hard”(p192) [16].

Limitations & Future Work

Limitations

- We evaluate only the TBox matching datasets that match classes, object properties, and datatype properties. ABox matching datasets (including individual data instances) are not considered due to privacy concerns in our targeted application domain.
- Due to the high cost of API calls for API-accessed commercial LLMs, experiments with the newest commercial models (e.g. OpenAI o1 and Anthropic claude-3-opus) are not included in this study.
- There may be additional resource requirements for running open-source LLMs locally. The run time for API-accessed commercial LLMs is controlled by the LLM providers.

Future Work

- Multimodal OM: We have packaged our system into several natural language-based commands. It could be integrated with advanced LLM functions to support multimodal input.
- Multilinguistic OM: Agent-OM supports ontologies in multiple languages. Results are not included here because there are few benchmarks.
- Small language model (SLM) for OM: SLM (e.g. gemma-2-2b) are useful in resource-constrained devices, but they have problematic tool interfaces at present.

Conclusion

Conclusion

- We introduce a new design paradigm for OM systems. Agent-OM, an agent-powered LLM-based framework, is proposed and implemented with a proof-of-concept system.
- The system has shown a powerful capability to perform OM tasks at different levels of complexity, leveraging the potential of using LLM agents for OM.
- It is a simple, lightweight, and natural language-driven approach with high scalability. Agent-OM is all you need.
- While OM has been studied for two decades or more, we are now at a point where the goal of 100% accurate, fully-automated, and domain-independent OM seems to be within reach.

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- Agent-OM did not participate in the OAEI 2022 and 2023 campaigns. According to the OAEI data policy (retrieved December 1, 2024), “OAEI results and datasets, are publicly available, but subject to a use policy similar to the one defined by NIST for TREC. These rules apply to anyone using these data.” Please find more details from the official website:
<https://oei.ontologymatching.org/doc/oei-deontology.2.html>.

Acknowledgements ii

- Agent-OM has been included in Professor Kerry Taylor's keynote speech in the International Workshop on Ontology Matching (OM) 2024 on November 13, 2024 (Baltimore, Maryland): "Leveraging Large Language Models for Ontology Matching". Please find more details from <https://om.ontologymatching.org/2024/>.
- This slide has been presented at the tabular data, semantics and AI seminar series on January 27, 2025 (IBM Research, Virtual).
- The theme is adapted from the Metropolis Beamer Theme. Metropolis was originally created by Matthias Vogelgesang and is available from <https://github.com/matze/mtheme> as well as CTAN. The theme *itself* is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.

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Thanks for your attention. Any questions?

- Agent-OM Full Paper: <https://www.vldb.org/pvldb/vol18/p516-qiangu.pdf>
- Agent-OM Official GitHub: <https://github.com/qzc438/ontology-llm>
- Agent-OM Discussion Group: <https://groups.google.com/g/agent-om>